

Section: **3.1 – STRENGTH OF METALLIC PARTS**
 Chapter: **3140**
 Title: **COLUMN BUCKLING IN COMPRESSION**
 Revision: **1.0**
 Date: **26/06/2019**

1 INTRODUCTION

1.1 OBJECT

This chapter is concerned with the buckling of uniform columns with compression loads applied.

1.2 METHOD APPLICABILITY

Analysis methodologies described in the present chapter is only applicable for columns which have the following characteristics:

- ✓ *Slender columns made of metallic materials.*
- ✓ Columns with *uniform cross-section* (constant cross-section shape along column's length).
- ✓ Columns with loads applied in *compression, uniform* (locally applied at column ends only), and *centred* (no secondary bending loads).

1.3 COLUMN STABILITY FAILURE MODES

Column's failure could occur in different manners, basically depending on the column slenderness ratio (relation between column length and its cross-section dimensions) and cross-section's stability. Instability under compression loads can be caused by column's *bending, bending/crippling interaction, torsion* or *combined bending/torsion*, which define the column's failure modes.

Failure of columns with stable cross-section will be usually caused by *primary instability* or *column buckling*, which is characterized by bending and/or torsion effects in the column, with displacements and/or rotations of the cross-section but without any modification of its initial shape. In the other hand, failure of columns with unstable cross-section will be usually caused by combination of a *primary and secondary instability*, usually interacting with crippling effects.

2 METHOD DESCRIPTION

The following paragraphs include methodologies for calculation of buckling critical compression stress for the different types of failure modes mentioned in §1.3. The minimum critical stress obtained for each one of those failure modes is the column's buckling allowable stress under compression loads.

Depending in column slenderness ratio, buckling could occur in elastic or plastic regime. Very slender columns rise the buckling load at a stress level below material's yield limit (buckling in elastic regime); in the other hand, in case of short columns, some areas/fibres of the column will enter in plastic regime before rising the buckling load (buckling in plastic regime).

2.1 BENDING PRIMARY INSTABILITY

Failure due to bending primary instability of a uniform column with stable cross-section occurs respect to the axis of minimum moment of inertia. General expression for critical buckling compression load P_F calculation, considering material's plasticity correction, is the one obtained by *Euler-Engesser*:

$$P_F = (C_F \pi^2 E_t I) / L^2$$

where:

C_F is the fixity coefficient for different column's supporting conditions (see Table 1).

E_t is the material's tangent modulus. If buckling occurs in elastic regime (buckling stress below column material's yield limit), $E_t = E_c$ (classical *Euler* column-buckling equation).

I is the minimum moment of inertia of column's cross-section.

L is the column's length.

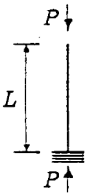
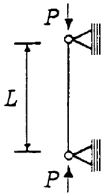
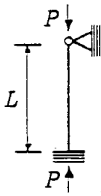
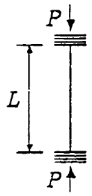
Column supporting conditions				
C_F	0.25	1.00	2.05	4.00

Table 1. Fixity coefficient C_F for different supporting conditions, §10.2 of Ref. 1.

Then, the critical buckling compression stress σ_F can be calculated as:

$$\sigma_F = P_F / A = (C_F \pi^2 E_t) / (L / r)^2 \rightarrow \text{Slenderness ratio} = L / (r \cdot C_F^{1/2})$$

where:

A is the area of column's cross-section.

r is the minimum radius of gyration of column's cross-section, calculated as: $r = (I / A)^{1/2}$

Notice that material tangent modulus E_t depends on σ_F value, $E_t = E_t(\sigma_F)$; therefore, an iterative process is needed in order to obtain σ_F .

2.2 BENDING PRIMARY INSTABILITY AND CRIPPLING INTERACTION

Compression failure of short columns with unstable cross-section (especially those with open cross-section with thin walls), will be probably caused by an interaction between bending primary instability and crippling. This kind of interaction can be analysed considering a modified version of *Johnson-Euler* approach, which traditional equation represents, in a failure stress σ_{FC} vs. slenderness ratio coordinate system, the following parabolic curve (Ref. 3, Chapter 7):

$$\sigma_{FC} = \sigma_{cc} \cdot [1 - (\sigma_{cc} (L / (r C_F^{1/2}))^2 / (4\pi^2 E_{to}))]^2$$

where (see §2.1 for definition of the rest of used parameters):

σ_{cc} is the crippling allowable stress of column's cross-section (see calculation method described in Chapter 3130 'Local buckling and Crippling of Beams/Stringers').

E_{to} is the material's tangent modulus at *Johnson-Euler* curve tangency point.

This expression is valid for $\sigma_{cc} / 2 \leq \sigma_{FC} \leq \sigma_{cc}$; therefore, columns which could fail due to this kind of interaction are the ones which complies with the following condition:

$$L / (r C_F^{1/2}) \leq [(\pi^2 E_{to}) / (\sigma_{cc} / 2)]^{1/2}$$

In case of columns with unusual high crippling allowable value, *Johnson-Euler* curve could derive in anti-conservative results since it could be, in some slenderness range, less critical than *Euler-Engesser* curve so, in such cases the minimum stress between both curves must be taken as the critical failure one.

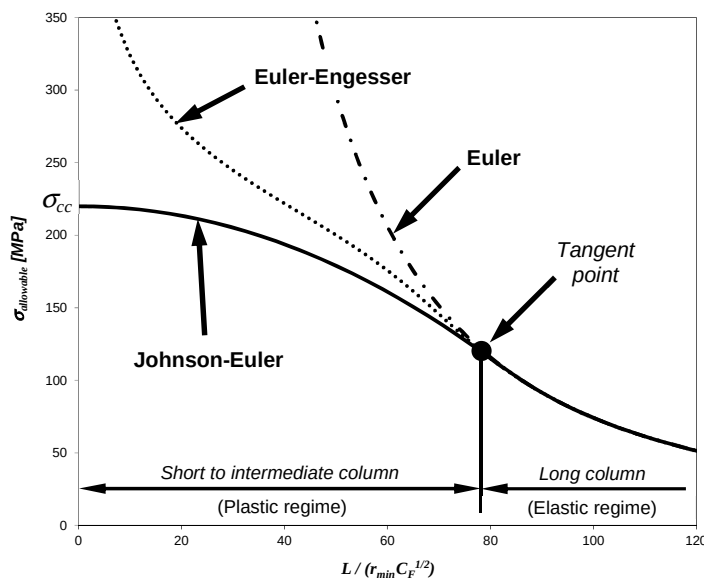


Figure 1. Example of buckling and buckling/crippling interaction failure modes curves comparison.

2.3 TORSION PRIMARY INSTABILITY

Compression failure of columns with cross-sections with low torsional stiffness, e.g. open sections with thin walls, will be probably caused by torsion instability or bending-torsion combined effect.

Based in sections 5.3 and 5.4 of Ref. 2, the general expression for calculation of critical buckling compression stress σ_T , caused by pure torsion instability, is the following one:

$$\sigma_T = (E_t / I_o) \cdot [(GJ / E_c) + (C_T \pi^2 C_w / L^2)]$$

where (see §2.1 for definition of the rest of used parameters):

- C_T is the fixity coefficient for different column's supporting conditions (see Table 2 or Ref. 2 for calculation procedure description for any supporting configuration).
- G is the material's shear elasticity modulus.
- E_c is the material's compression elasticity modulus.
- I_o is the polar moment of inertia of column's cross-section respect to its shear centre (see Figure 2).
- J is the torsional constant of column's cross-section.
- C_w is the warping constant of column's cross-section (see Chapter 3110 'Beam sections' and Table 10.2 of Ref. 4).

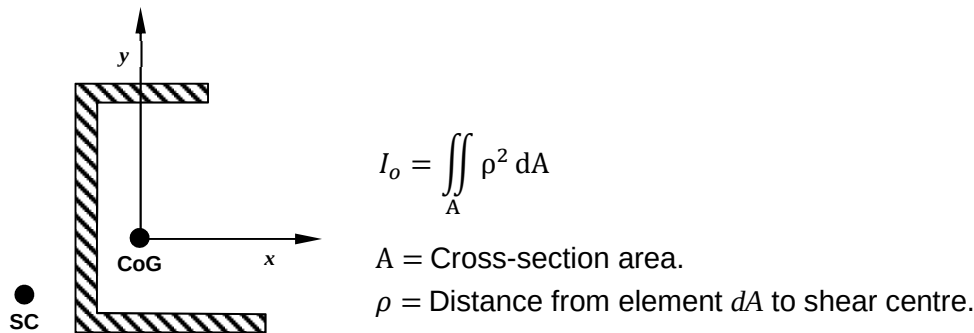


Figure 2. Calculation scheme of column's cross-section polar moment of inertia.

Supporting cond.	C_T
Rotation: Restricted in both ends of column. Warping: Allowed in both ends of column.	1
Rotation: Restricted in both ends of column. Warping: Restricted in both ends of column.	4
Rotation: Restricted in both ends of column. Warping: Allowed in one end of column and restricted in the other one.	2.05

Table 2. Fixity coefficient C_T for different supporting conditions.

If buckling occurs in elastic regime (buckling stress below column material's yield limit), $E_t = E_c$. Guidelines for calculation of column cross-section's torsional constant J and warping constant C_w can

be found in Chapter 3110 'Beams sections'. Notice that value of material tangent modulus E_t depends on σ_T value, $E_t = E_t(\sigma_T)$; therefore, an iterative process is needed in order to obtain σ_T .

2.4 COMBINED BENDING AND TORSION PRIMARY INSTABILITY

2.4.1 COLUMNS WITH UNSYMMETRICAL CROSS-SECTION

Based in section 5.5 of Ref. 2, the general equation for calculation of critical buckling compression stress σ_{FT} , caused by combined bending-torsion primary instability is the following one:

$$(I_o / A) \cdot [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_F / E_{tF})_x] \cdot [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_F / E_{tF})_y] \cdot [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_T / E_{tT})] - y_o^2 [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_F / E_{tF})_x] - x_o^2 [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_F / E_{tF})_y] = 0$$

where, in line with previous sections notation: $E_{tFT} = E_t(\sigma_{FT})$, $E_{tF} = E_t(\sigma_F)$ and $E_{tT} = E_t(\sigma_T)$ and ratios $(\sigma_F / E_{tF})_x$ and $(\sigma_F / E_{tF})_y$ correspond with pure bending buckling critical compression stress respect to principal axes OX and OY of column's cross-section (see Figure 3) so, using Euler's formula described in §2.1:

$$(\sigma_F / E_{tF})_x = (C_F \pi^2) / (L / r_x)^2 \quad ; \quad (\sigma_F / E_{tF})_y = (C_F \pi^2) / (L / r_y)^2$$

where r_x and r_y are the minimum radii of gyration of column's cross-section around principal axes OX and OY, respectively.

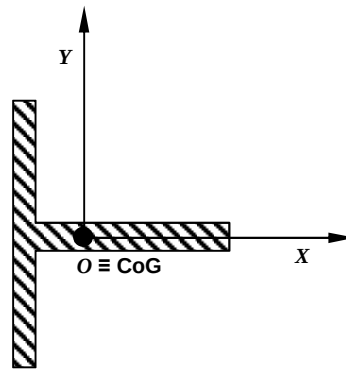


Figure 3. Column typical cross-section reference axes.

Then, ratio (σ_T / E_{tT}) corresponds with the pure torsion buckling critical stress of the column's so, using §2.3 expression:

$$(\sigma_T / E_{tT}) = I_o^{-1} \cdot [(GJ / E_c) + (C_T \pi^2 C_w / L^2)]$$

According to this, critical buckling compression stress σ_{FT} can be obtained following these steps:

- 1.- Ratios $(\sigma_F / E_{tF})_x$, $(\sigma_F / E_{tF})_y$ and (σ_T / E_{tT}) are calculated with above mentioned expressions.
- 2.- General equation is solved considering the ratio (E_{tFT} / σ_{FT}) as the incognita, obtaining three different equation roots α .

- 3.- The highest root (α_{max}) will result in the minimum allowable value so, it must be selected and used to solve the following equation:

$$E_{tFT} / \sigma_{FT} = \alpha_{max}$$

Material tangent modulus E_{tFT} depends on σ_{FT} value, $E_{tFT} = E_t(\sigma_{FT})$; therefore, an iterative process is needed in order to obtain σ_{FT} .

2.4.2 COLUMNS WITH SINGLE-SYMMETRY CROSS-SECTION

In this case, shear centre is located at the symmetry axis so, no torsion effect is coupled with buckling due to bending around the axis perpendicular to the symmetry one. Therefore, the possible buckling failure modes for this kind of column are the following ones:

- Bending primary instability respect to the cross-section's axis perpendicular to the symmetry one.
- Combined bending-torsion primary instability respect to the cross-section's symmetry axis.

For example, if OX is the cross-section symmetry axis (see Figure 3), column's buckling critical stress will be the minimum one of the following stresses:

- Buckling critical compression stress due to bending around cross-section's axis OY, $(\sigma_F)_y$, obtained using the Euler's formula (see §2.1).
- The minimum value of roots σ_{FT} of the following second order equation, which can be solved in a similar way as the third order's general equation:

$$(I_o / A) \cdot [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_F / E_{tF})_x] \cdot [1 - (E_{tFT} / \sigma_{FT}) \cdot (\sigma_T / E_{tT})] - x_o^2 = 0$$

An analogous method should be followed if OY is the cross-section symmetry axis.

2.4.3 COLUMNS WITH DOUBLE-SYMMETRY CROSS-SECTION

If shear centre of column's cross-section is the same as its centre of gravity (case of two symmetry axes), it can be considered that there is no interaction between bending and torsion failure modes. Therefore, column's buckling critical compression stress will be the minimum among the values obtained for bending and for torsion instabilities.

3 APPENDICES

ABBREVIATIONS AND SYMBOLS

<i>Symbol</i>	<i>Units</i>	<i>Definition</i>
A	mm^2	Area of column's cross-section
C_F, C_T	-	Fixity coefficients of column
C_w	mm^6	Warping constant of column's cross-section
E	MPa	Material's elasticity modulus
E_c	MPa	Material's compression elasticity modulus
E_t	MPa	Material's tangent modulus
E_{to}	MPa	Material's tangent modulus at <i>Johnson-Euler</i> curve tangent point
G	MPa	Material's shear elasticity modulus
I	mm^4	Minimum moment of inertia of column's cross-section
I_o	mm^4	Polar moment of inertia of column's cross-section respect to its shear centre
J	-	Torsional constant of column's cross-section
L	mm	Column's length
P_F	N	Critical buckling failure load
r	mm	Minimum radius of gyration of column's cross-section
r_x, r_y	mm	Radii of gyration of column's cross-section around principal axis OX or OY
σ	MPa	Generic failure axial stress
σ_{cc}	MPa	Crippling allowable stress of column's cross-section
σ_F	MPa	Critical bending primary instability failure compression stress
σ_{FC}	MPa	Critical bending/crippling interaction instability failure compression stress
σ_{FT}	MPa	Critical bending/torsion interaction instability failure compression stress
σ_T	MPa	Critical torsion primary instability failure compression stress
x_o, y_o	mm	Column's cross-section shear centre coordinate respect to principal axis OX or OY

REFERENCES

	<i>Document Reference</i>	<i>Issue</i>	<i>Title</i>
Ref. 1	[M. C. Y. Niu]	2 nd Ed.	"Airframe Stress Analysis and Sizing"
Ref. 2	[S. P. Timoshenko & J. M. Gere]	2 nd Ed.	"Theory of Elastic Stability"
Ref. 3	[E. F. Bruhn, B. S., M. S., C. E.]	1973	"Analysis and Design of Flight Vehicle Structures"
Ref. 4	[W. C. Young & R. G. Budynas]	7 nd Ed.	"Roark's Formulas for Stress and Strain"

SOFTWARE

<i>Program</i>	<i>Version</i>	<i>Description</i>
STRENGTH2000 (Airbus DS)	TBC	Detailed sizing of metallic and composite aircraft structures
PANCO (Airbus DS)	4.2	This program analyses the column-buckling of variable cross-section beams and performs the beam-column strength checks for different column's supporting conditions.

Table 3. *Software programs referenced in this chapter*

DISTRIBUTION LIST

<i>Department</i>	<i>Name</i>	<i>Full Doc.</i>	<i>Cover Page</i>
Stress Methods & Optimisation (TEASP-TL3)	Dr. Fernaß Daoud	X	
Stress Methods & Optimisation (TEASP-TL3)	Dr. Alexander Hörmann	X	
Stress M&L, Derivatives, Mission (TEASA-TL2)	Miguel Angel Calero Gallardo	X	
Stress Combat MAN (TEASA-TL3)	Sebastian Rönz	X	
Stress Combat, Aerostructures, A400M and Technologies GET (TEASA-TL4)	Sofia Ponce Borrero	X	
Structural Repair GET (TEASA-TL6)	Jose Luis Melchor Martin de la Sierra	X	

Table 4. *Distribution list (of electronic document, ahead of public release)*

RECORD OF REVISIONS

<i>Revision Date</i>	<i>Authors</i>	<i>Change Reason</i>	<i>Pages/Sub-chapters affected</i>
0.4 – 1.0 11/10/2018	Guillermo Moreda	First issue	All pages

Table 5. *Record of Revisions: authors, change reasons and sections/pages affected*

APPROVAL SIGNATURES TABLE

<i>Dpt. \ Site</i>	<i>Getafe</i>	<i>Manching</i>
Stress Analysis (TEASA)	Miguel Ángel Calero 26/06/2019	Bernhard Mayer 26/06/2019
Stress Methods (TEASP-TL3)	G. Baños 26/06/2019	Florian Glock 26/06/2019

Table 6. *Approval signatures table for last revision of this chapter*